

# Designing a Digital Game Simulation about Collaborative Working Skill for Young Millennials.

*Nurizka Fitrianingrum*<sup>1,\*</sup>, *Dianing Ratri*<sup>1</sup>, and *Wawan Dhewanto*<sup>2</sup>

<sup>1</sup>Institut Teknologi Bandung, Faculty of Art and Design, Bandung, Indonesia

<sup>2</sup>Institut Teknologi Bandung, School of Business and Administration, Bandung, Indonesia

**Abstract.** Young millennials have the potential to become the new leader in the future workforce, but they have a tendency to choose work alone. In order to nurture their potential, they need to improve their collaboration skill so they could grow in a workplace with good collaborative work culture. Qualitative approach with design thinking framework used in this research to produce a game design that will help young millennials to habituate collaboration skill. Those habituate process could be happening within a game that simulates collaborative work pattern, played with more than two players, and give them a chaotic situation that lead them to communicate and solving the problem together.

## 1 Introduction

Young millennials are the generation that already enter young adult phase with age of 20-28 years old in 2020 (born in 1992~2000) [1-4]. This generation are the new light of workforce. They are expected to be the new leader in workforce, especially because they are really suitable to deal with the fast-forward digital technologies innovation since they are digital natives, they are growing up with digital technologies itself [4].

Young millennials need to improve their collaboration skill so they could grow in workplace with collaborative work culture. Those workplace with collaborative work culture will help young millennials to nurture their capabilities and potential [5]. The purposes of this research are to design a game that could simulate collaborative working skill. That game will help young millennials to habituate collaboration skill in a safe space where they could try and no need to worry about failure.

## 2 Design Process

This research design process will follow qualitative approach with design thinking framework. Design process will be divided into five phases: empathise, define, ideate, prototyping and testing. All elaboration for each phase will elaborated below.

---

\* Corresponding author : nurizkaf96@students.itb.ac.id

2.1 Empathise

The problems in young millennials regarding collaborative working culture are they have expectation for their work colleagues to work collaboratively [6] but they have tendency to choose work alone and just rely on their own abilities [4]. Also, they are experiencing bad stigma from older generation as they called to be entitled, lazy, egoistic, and shallow [5]. They understand they need to improve their abilities and capabilities [2], but in order to grow in workplace that have a good collaborative working culture they need to improve their collaboration skill so they could be nurtured better and learn other skill better.

2.2 Define

First thing before deciding how to habituate collaboration skill we need to understand that collaborative work have specific elements in order to be called collaborative work [7–9]. Those elements will help us to expand collaborative learning concept [10, 11] so it could help to create collaborative working game.

Collaborative learning is a concept where there is a team with each member have roles and doing activities or tackle problems together with same objectives and goals. Those situations eventually will lead to chaos situation that force members to communicate with each other, so they could actively find problem patterns and solution [11].

To implement chaos situation in collaborative learning, we need to simulate it on game became simulation on game will help us to easily represent those complicated situations [12, 13] and give player a space to freely explore and having no worries to fail [13].

There is implementation of collaborative learning on game framework that based on collaborative interaction and multiplayer game. This resulted in nine patterns in four category called collaborative game pattern [10].

As from existing game, there are three notable games that represent multiplayer game with collaborative element but with different kind of mechanism as it shows in Table 1.

Table 1. Point taken from existing multiplayer game with collaborative element

| Name                           | Description                                                                                                                              | Point Taken                                                                                                                                                                                                                                                 |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forbidden<br>Dessert           | Digital board game, with four to six players in one screen device gameplay                                                               | Equal roles, where all players have to contribute and all players are as important as other. To the point the game will be over if one player dies                                                                                                          |
| Overcooked                     | Shared screen multiplayer game with three to four players in one screen console with different controller                                | This game didn't assign specific roles to each player but it level design will force players to assign their roles instinctively also this game design it level to create chaos situation and that situation will lead people to communicate more           |
| Ragnarok<br>M: Eternal<br>Love | Massively multiplayer online role-playing game with up to twelve players on two team in dungeon run (especially Thanatos Tower Dungeon). | Gives player freedom to choose their job and how to build it. However too much players, imbalance on power, and complicated game mechanics lead to some case where some player didn't contribute as much as other even nearly became an obstacle in dungeon |

From all examples above, we could conclude that 1) the game need to give common goals for uniting the players, 2) every players get different roles and abilities but not to complex so it will open up multiple opportunity to finish the game, 3) game should have chaos element that will force players to communicate and those element enriched with tension resources, 4) game should have different approaches to finish it so players will

directed to solving problem not just found a way, 5) game should have level completion requirement that need every players to do something so there is no player that feel unneeded.

2.3 Ideate

From conclusion in define phase, we need to choose what kind of genre and general gameplay mechanism that suitable to fulfil all the points. Puzzle simulation with shared-screen multiplayer will be the most suitable. First, because puzzle would give player a simplistic mechanism not too overwhelming on the front like most role-playing out there but puzzle still give much room to trigger creative problem solving [13]. Second, because shared screen multiplayer would give player a sense that they are play it together with same point of view.

To implement puzzle simulation that fit with young millennials sentiment we need to combine with optimism theme such saving the earth, because this generation have big concern in sustainability and climate change [3]. Elaboration from combining between saving earth theme, puzzle simulation, shared-screen multiplayer with collaborative game pattern as the main corridor could be seen on Table 2.

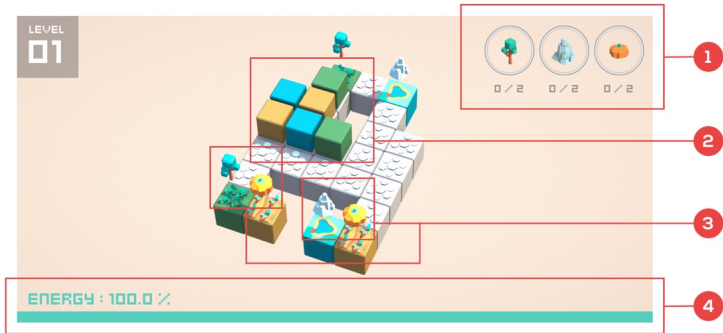
Table 2. Game Mechanics and Concept Implementation.

| No | Component                 | Elaboration/Implementation                                                                                                                                                                                                                                                                                                                                                                     |
|----|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Purpose                   |                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Common Objectives         | Restoring earth condition with collecting earthy resources (plant, water and food)                                                                                                                                                                                                                                                                                                             |
|    | Win Condition             | Required earthy resources already fulfilled.                                                                                                                                                                                                                                                                                                                                                   |
|    | Lose Condition            | Players owned resources for surviving together already depleted.                                                                                                                                                                                                                                                                                                                               |
| 2  | Players                   |                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Number of                 | Three.                                                                                                                                                                                                                                                                                                                                                                                         |
|    | Player Roles              | Player will be assigned to different roles. 1) Ecologist collect Plant Resources, 2) Hydrologist collect Water Resources, 3) Agriculturist collect Food Resources.                                                                                                                                                                                                                             |
| 3  | Collaborative Task        |                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Concurrency Pattern       | Players will move their characters on puzzle platform. Each player will move one or more character pion. Each character pion could block other character so they could plan to find the most efficient and effective movement to complete their game.                                                                                                                                          |
|    | Parallelization Pattern   | The concurrency pattern (pattern that will create a task that could only done if two or more player doing something with their corresponding roles and abilities) and the parallelization pattern (pattern that will create task that could be done with dispersing the players so the task could be done faster) will happen in accordance with the resources position and player creativity. |
| 4  | Resources                 |                                                                                                                                                                                                                                                                                                                                                                                                |
|    | Relate to game objectives | Collectible resources as earthy resources (plant, water, and food). Obstacle as portal to teleport, button to move the puzzle platform, button to switch position.                                                                                                                                                                                                                             |

| No | Component                          | Elaboration/Implementation                                                                                                                                              |
|----|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Relate to game tension             | Energy resource as shared resources that will be depleted every move player made. Time counter as resources to show how long they are playing and affected group score. |
|    | Relate to supporting player        | Energy orb to increased energy resources so player could play longer and better.                                                                                        |
| 5  | Interaction Process                |                                                                                                                                                                         |
|    | Out-game (happen outside the game) | Interaction such give another player information and coordinating directly to move the characters.                                                                      |
|    | In-game                            | Interaction that happen between player and game system such hint and tutorial.                                                                                          |

### 2.4 Prototyping

All the concept elaborated in ideate phase translated into game prototype as seen in Fig 1. Those are the earliest level where players will just be collecting earthy resources with limited energy resources without any complicated obstacle or helpful energy orbs. Characters and their corresponding earthy resources have same colour coded. Colour code for characters and earthy resources as following: a) Ecologist-Plant with their green colour, b) Hydrologist-Water with blue colour, c) Agriculturist-Food with yellow colour.



**Fig. 1.** Game Interface. Red mark with number indicates game elements. (1) Progress bar indicate how much earthy resources need to collected in order clearing the game and already collected while progressing the game. (2) Characters pion, each color indicate one characters and each pion have linked movement. (3) Earthy resources on stage that need to be collected. (4) Energy bar that will depleted each time players moves the characters.

### 2.5 Testing

The game was tested with short version of User Experience Questionnaire (UEQ-S) [14] method to understand their overall experiences expectation. After that, we use focus group discussion discussing about their feedback about unique point, their experience about shared-screen multiplayer, nuisance in their play experiences, and about what need to improve. The game that used in this test was Level 01 where the level consists only of basic resources and mechanism. The test was conducted with three participants with age range between 24-26 on Monday, 8<sup>th</sup> June 2020. All participants played the game as one team and try different state of resources to find how much resources needed so player could play it properly.

The result of the test was positive, it shows that this game fulfils participants experience expectations as the overall UEQ-S score is 2,083 (2,333 for pragmatic quality and 1,833 for hedonic quality). Also, participants give positive feedback on focus group discussion such they feel the game give them room to collaborate and communicate with their friends and the visual was so good, but they also state this game still need to improve such give proper tutorial and give more obstacles and challenges.

### 3 Conclusion

This research shows the potential of puzzle simulation with shared-screen multiplayer game with implementation of collaborative game pattern could simulate collaborative working skill for young millennials. This research still needs to improve as it need to be exposed to more test participants and improving with more complicated game mechanics.

### References

1. R. J. Bonnie, C. Stroud, and H. Breiner, *Investing in the health and well-being of young adults* (National Academies Press, 2015)
2. Deloitte Global, *Deloitte Global Millennial Survey 2019* (2019)
3. Deloitte Global, *Deloitte Global Millennial Survey 2020* (2020)
4. Ernst & Young, *Next-gen workforce: secret weapon or biggest challenge?* (2019)
5. IBM, *Myths, exaggerations and uncomfortable truths. The real story about Millenials in workplace* (2015)
6. Randstad, *Gen Z and Millennials Collide at Work* (2016)
7. H. Patel, M. Pettitt, and J. R. Wilson, *Factors of collaborative working: A framework for a collaboration model*, *Applied Ergonomics*, **43**(1), 1–26 (2012)
8. H. Getha-Taylor, *Identifying Collaborative Competencies*, *Review of Public Personnel Administration*, **28**(2), 103–119 (2008)
9. G. Anderson, *Mastering Collaboration : Make Working Together Less Painful and More Productive, 1st Edition* (Sebastopol, O'Reilly Media, 2019)
10. C. Reuter, V. Wendel, S. Göbel, and R. Steinmetz, *Game Design Patterns for Collaborative Player Interactions*, *DiGRA 2014*, **8**, (2014)
11. S. Laisema and P. Wannapiroon, *Collaborative Learning Model with Virtual Team in Ubiquitous Learning Environment using Creative Problem-Solving Process*, *International Journal on Integrating Technology in Education*, **2**(4), 1–14, (2013)
12. D. Duchatelet, D. Gijbels, P. Bursens, V. Donche, and P. Spooren, *Looking at role-play simulations of political decision-making in higher education through a contextual lens: A state-of-the-art*, *Educational Research Review*, **27**, 126–139 (2019)
13. J. Novak, *Game Development Essentials: An Introduction* (Cengage Learning, Delmar, 2012)
14. M. Schrepp, A. Hinderks, and J. Thomaschewski, *Design and Evaluation of a Short Version of the User Experience Questionnaire (UEQ-S)*, *International Journal of Interactive Multimedia and Artificial Intelligence*, **4**(6), 103, (2017)